Bounded Communication Fault Tolerance

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Introduction

Today:

- Bounded Communication Fault Tolerance
  1. Walktrough the math
  2. Alternative definition to Expander decomposition
  3. (d, k) - Toric
- New experimental results.
- Answering the code teleportation.

Bounded Communication Fault Tolerance

Setting

We allow a non-nosy, strong as we wish, classical computation aside. Yet we would like to limit the communication to a constant number of bits per logical qubit, in each round.

Justification

1. If we don't have a constant overhead in depth, in that model, $\Rightarrow$ we also don't have it in the standard model.

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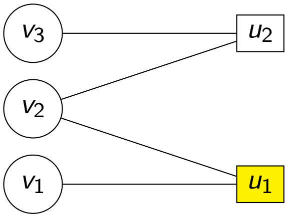
2. In that model, gate teleportation for a constant dimension code is easy. Namely, for the Topological codes the task is reduced to finding a decoding with a constant-size hint.

Max Color

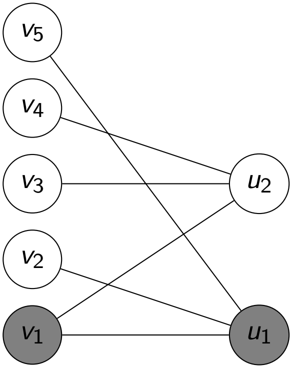
$$
\begin{tikzcd}
\text{Given - } \rho \arrow[r] \arrow[d] & \text{Decoding} \arrow[r] \arrow[d] & p\text{-depololraized} \arrow[d] \\
E_1 & E_2 & E_3
\end{tikzcd}
$$

Figure: Illustration of the cycle.

standard model, $t = 0$



simplified (p', q)-model



SWIFT

What did we learn?

1. Storing time is, likily, not $\Theta(e^n)$, might be $\Theta(L^\alpha)$ for $\alpha \in (0, 1)$.
2. $\Rightarrow$ Implies that the structure of the theorem/key-lemma is different than the known refrigerators.

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   - More experiments, higher statistical standards?
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3. How to use the fact our code is a quotient code?